Figure 1.2 A snapshot of the Ethereum public blockchain

Figure 1.2 also shows blocks of Txs. Each block (Bk) is made up of a set of transactions and is identified by a block number. Block #10163275 has 142 Txs, and block #10163274 has 60 Txs. You may see a different set of blocks when you visit the site. But you can always search for a particular block number (#10163275, in this case) and verify the number of Txs. The block will have *the same number of Txs* shown here, exemplifying the immutable nature of blockchain technology. The blocks are linked to form a chain of blocks, or blockchain.

1.2 What is a blockchain?

A *blockchain* is a technology for enabling trust in a decentralized system of transacting peer participants. The purpose of a blockchain is to verify and validate (or reject, if not valid) a transaction initiated by a participant, and then execute the transaction and record the proof of these actions with the consensus of the peer participants. As shown in figure 1.3, the blockchain-based trust infrastructure exists within a larger system. Blockchain infrastructure contains software for a specific purpose: trust intermediation among a large number of (typically unknown) peer-to-peer participants. On the left side of figure 1.3 is a distributed (client/server) system performing routine operations. This system may send messages that contain data to be verified, validated, and recorded on the blockchain (on the right) to